

Misurazione Implicita in Psicologia: **L'analisi classica dei dati IAT**

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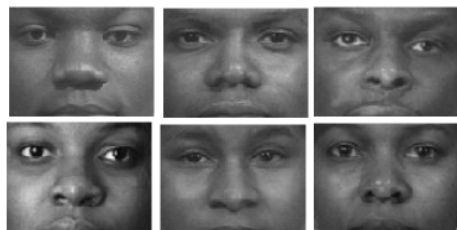
Implicit Association Test

12 stimoli target

White people faces



Black people faces



16 stimoli attributo

Attributi positivi

Attributi negativi

Good, laughter, pleasure, glory, peace, happy, joy, love

Evil, bad, horrible, terrible, nasty, pain, failure, hate

Scoring

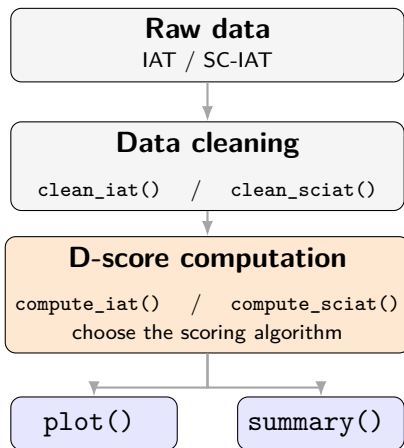
Person-level scores

$$s_p = \frac{\bar{X}_{p,\text{comp}} - \bar{X}_{p,\text{inc}}}{sd_{\text{pooled}}}$$

<i>D</i> score	Risposte errate	Risposte troppo veloci
<i>D1</i>	Built-in correction	No
<i>D2</i>	Built-in correction	Eliminare trials < 400 <i>ms</i>
<i>D3</i>	Mean (correct responses) + 2 <i>sd</i>	No
<i>D4</i>	Mean (correct responses) + 600 <i>ms</i>	No
<i>D5</i>	Mean (correct responses) + 2 <i>sd</i>	Eliminare trials < 400 <i>ms</i>
<i>D6</i>	Mean (correct responses) + 600 <i>ms</i>	Eliminare trials < 400 <i>ms</i>

Workflow

```
install.packages("implicitMeasures")  
library(implicitMeasures)
```



```
data("raw_data") # load data
str(raw_data)
```

```
'data.frame':    84726 obs. of  6 variables:
 $ Participant: int   4 4 4 4 4 4 4 4 4 4 ...
 $ latency    : int  2592 628 808 783 2059 1114 608 663 771 676 ...
 $ correct    : int   1 1 1 1 1 1 1 1 1 1 ...
 $ trialcode  : Factor w/ 32 levels "age","alert",...: 31 5 3 20 3 20
 $ blockcode  : Factor w/ 13 levels "demo","practice.iat.Milkbad",...
 $ response   : Factor w/ 46 levels "", "0", "1", "19",...: 43 43 43 43
```



```
iat_cleandata <- clean_iat(raw_data, sbj_id = "Participant",  
                           block_id = "blockcode",  
                           mapA_practice = "practice.iat.Milkbad",  
                           mapA_test = "test.iat.Milkbad",  
                           mapB_practice = "practice.iat.Milkgood",  
                           mapB_test = "test.iat.Milkgood",  
                           latency_id = "latency",  
                           accuracy_id = "correct",  
                           trial_id = "trialcode",  
                           trial_eliminate = c("reminder", "reminder1"),  
                           demo_id = "blockcode",  
                           trial_demo = "demo")
```

```
str(iat_cleandata)
```

List of 3

```
$ data_keep      :Classes 'iat_clean' and 'data.frame': 19440 obs. of  8 variables:
..$ participant   : int  [1:19440] 4 4 4 4 4 4 4 4 4 4 ...
..$ latency       : int  [1:19440] 1282 1299 1435 1089 967 648 967 615 729 642 ...
..$ correct       : int  [1:19440] 1 1 1 1 1 1 1 1 1 1 ...
..$ block_original: chr  [1:19440] "practice.iat.Milkbad" "practice.iat.Milkbad" ...
..$ condition     : chr  [1:19440] "MappingA" "MappingA" "MappingA" "MappingA" ...
..$ block_pool    : chr  [1:19440] "practice" "practice" "practice" "practice" ...
..$ block         : chr  [1:19440] "practice_MappingA" "practice_MappingA" "practice_MappingA" ...
..$ trial_id      : Factor w/ 32 levels "age","alert",...: 20 6 20 6 20 6 20 6 20 6 ...

$ data_eliminate:'data.frame': 64638 obs. of  6 variables:
..$ Participant: int  [1:64638] 4 4 4 4 4 4 4 4 4 4 ...
..$ latency     : int  [1:64638] 2592 628 808 783 2059 1114 608 663 771 676 ...
..$ correct     : int  [1:64638] 1 1 1 1 1 1 1 1 1 1 ...
..$ trialcode   : Factor w/ 32 levels "age","alert",...: 31 5 3 20 3 20 5 3 20 3 ...
..$ blockcode   : Factor w/ 13 levels "demo","practice.iat.Milkbad",...: 4 4 4 4 4 4 4 4 4 4 ...
..$ response    : Factor w/ 46 levels "", "0", "1", "19",...: 43 43 43 43 43 43 43 43 43 43 ...

$ demo          :'data.frame': 3726 obs. of  6 variables:
..$ participant: int  [1:3726] 4 4 4 4 4 4 4 4 4 4 ...
..$ latency     : int  [1:3726] 53047 53047 53047 53047 21554 21554 11266 11266 11266 11266 ...
..$ correct     : int  [1:3726] 1 1 1 1 1 1 1 1 1 1 ...
```

```
dscore = compute_iat(iat_cleandata, Dscore = "d1")
summary(dscore)
```

\$general

measure value

1 Participants 162
2 Total trials 19440

\$statistics

	measure	mean	sd
1	Response time Mapping A	972.3939630	259.23858664
2	Response time Mapping B	752.5932639	196.28708624
3	Accuracy Mapping A	0.9316592	0.05958240
4	Accuracy Mapping B	0.9655245	0.05919172
5	D-score D1	-0.5271882	0.50846375

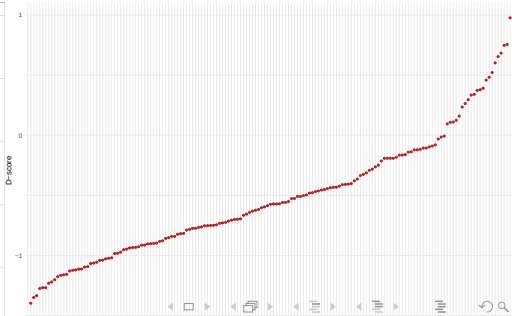
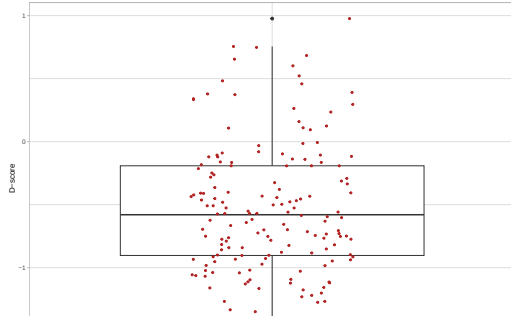
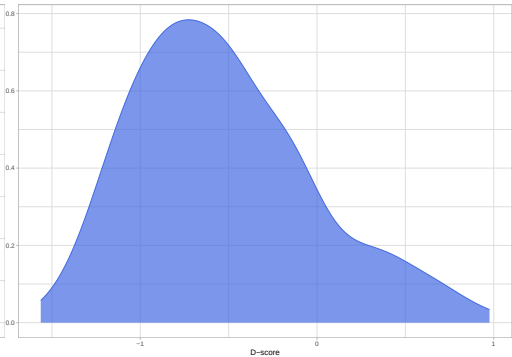
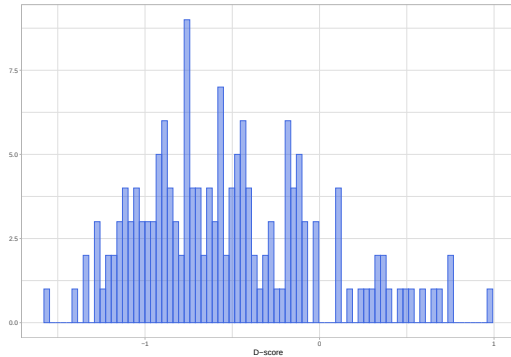
\$algorithm

[1] "D1"

attr(,"class")

[1] "summary_dscore" "list"

```
plot(dscore, graph = "histogram")  
plot(dscore, graph = "density")  
plot(dscore, graph = "boxplot")  
plot(dscore, graph = "points", order_sbj = "D-increasing")
```

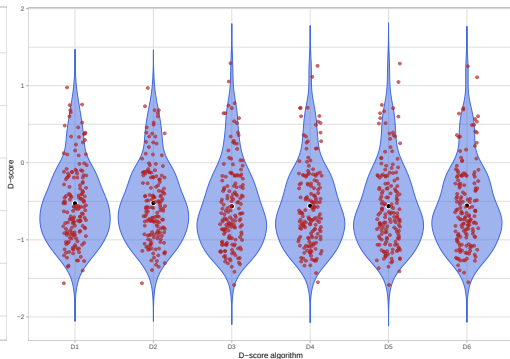
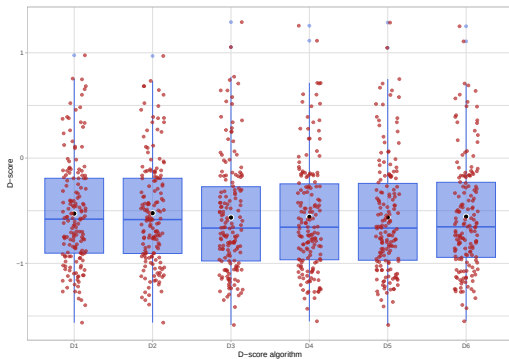


```
multi_iat = multi_dscore(iat_cleandata$data_keep)
str(multi_iat)
```

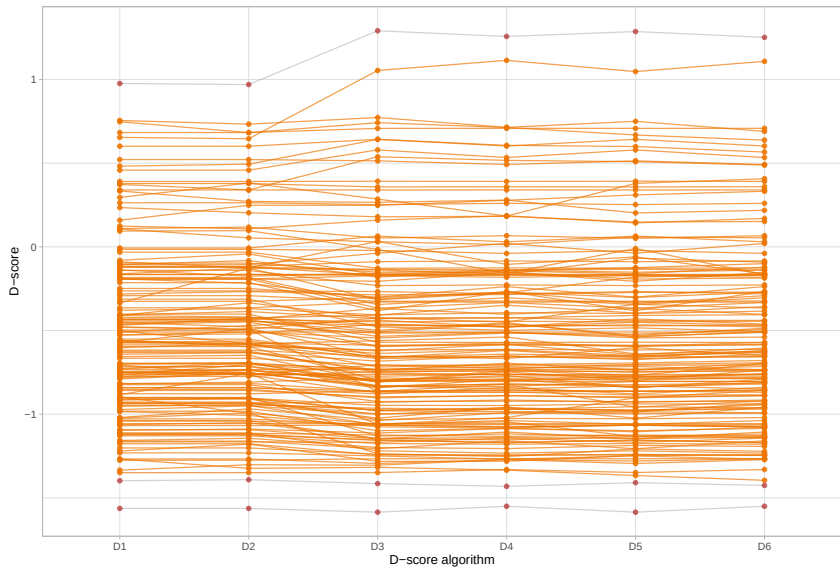
List of 4

```
$ scores      : 'data.frame':   162 obs. of  7 variables:
 ..$ participant: int [1:162] 4 6 8 11 14 17 18 19 20 21 ...
 ..$ dscore_d1  : num [1:162] -0.762 -1.231 -0.107 -0.953 -0.464 ...
 ..$ dscore_d2  : num [1:162] -0.762 -1.231 -0.107 -0.944 -0.449 ...
 ..$ dscore_d3  : num [1:162] -0.762 -1.25 -0.038 -0.965 -0.593 ...
 ..$ dscore_d4  : num [1:162] -0.7624 -1.2691 0.0202 -0.9721 -0.5739 ...
 ..$ dscore_d5  : num [1:162] -0.762 -1.25 -0.038 -0.955 -0.592 ...
 ..$ dscore_d6  : num [1:162] -0.7624 -1.2691 0.0202 -0.9629 -0.5721 ...
$ scores_long: 'data.frame':   972 obs. of  3 variables:
 ..$ participant: int [1:972] 4 6 8 11 14 17 18 19 20 21 ...
 ..$ score_type : Factor w/ 6 levels "D1","D2","D3",...: 1 1 1 1 1 1 1 1 1 1 ...
 ..$ score      : num [1:972] -0.762 -1.231 -0.107 -0.953 -0.464 ...
$ source      : chr "IAT"
$ labels      : chr [1:6] "D1" "D2" "D3" "D4" ...
- attr(*, "class")= chr [1:2] "multi_dscore" "list"
```

```
plot(multi_iat)
plot(multi_iat, graph = "violin")
plot(multi_iat, graph = "individual")
```



159 participants involved in at least one rank inversion



Prendete i dati disponibili sul [sito](#) del corso e importateli in R:

```
dati = read.delim("data/iat_data.dat")  
str(dati)
```

```
'data.frame':   16128 obs. of  7 variables:  
 $ id          : int   1 1 1 1 1 1 1 1 1 1 ...  
 $ order       : int   1 1 1 1 1 1 1 1 1 1 ...  
 $ test        : int   1 1 1 1 1 1 1 1 1 1 ...  
 $ blocknum    : int   3 3 3 3 3 3 3 3 3 3 ...  
 $ trialnum    : int   1 2 3 4 5 6 7 8 9 10 ...  
 $ correct     : int   1 1 1 1 1 1 1 1 1 1 ...  
 $ latency     : int  812 687 723 619 657 580 596 687 618 936 ...
```

Sono dati di uno IAT usato nel paper

Röhner, J., & Thoss, P. J. (2019). A tutorial on how to compute traditional IAT effects with R. *The Quantitative Methods for Psychology*, 15(2), 134–147.

e sono accessibili su OSF

Non si hanno i nomi dei blocchi, ma solo

```
table(dati$blocknum)
```

```
      3      4      6      7  
4032 4032 4032 4032
```

I blocchi 3 e 4 saranno la condizione A e i blocchi 6 e 7 la condizione B

Calcolare i punteggi IAT con il pacchetto `implicitMeasures` e investigare le possibili differenze attribuibili alla variabile `order`